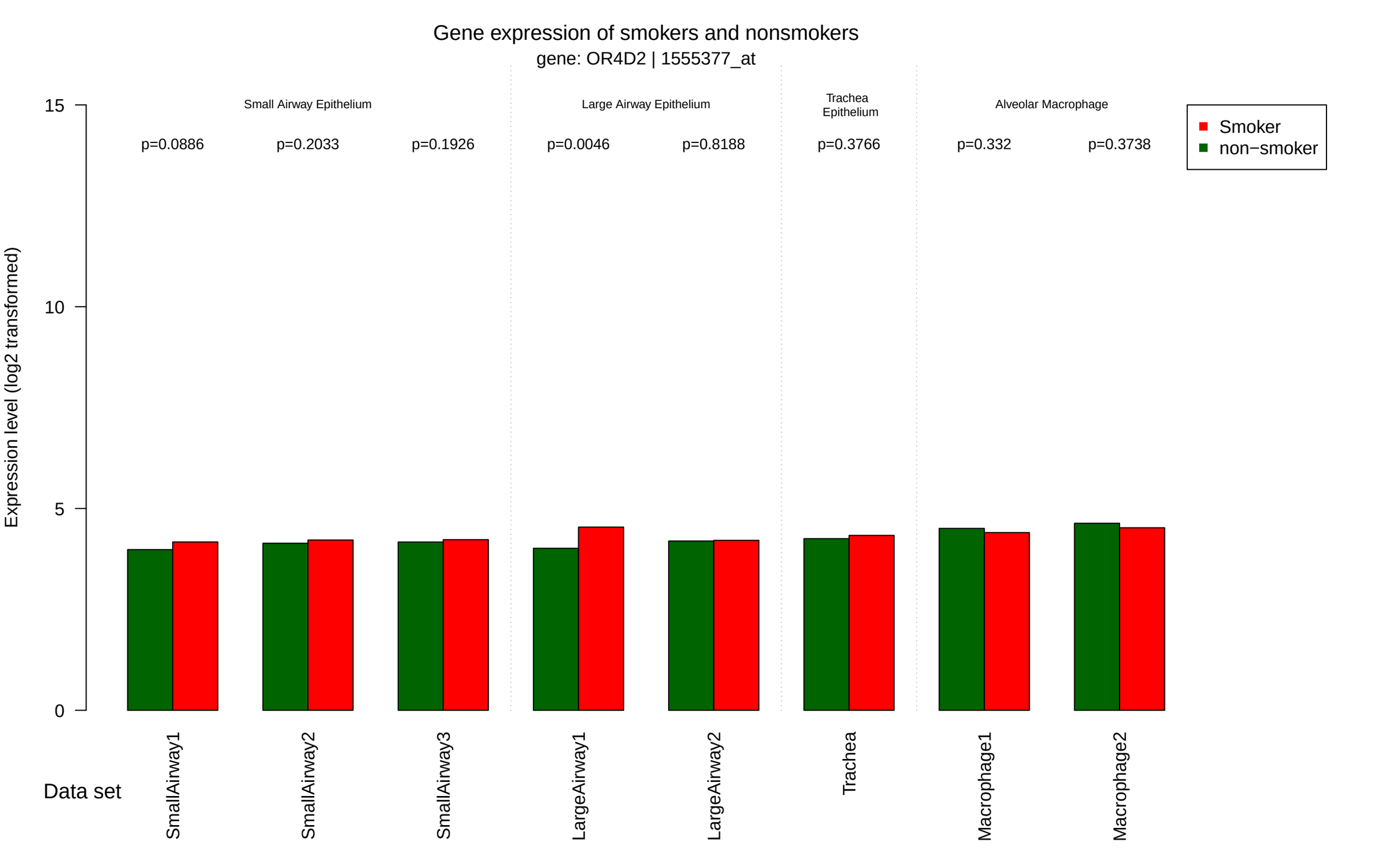
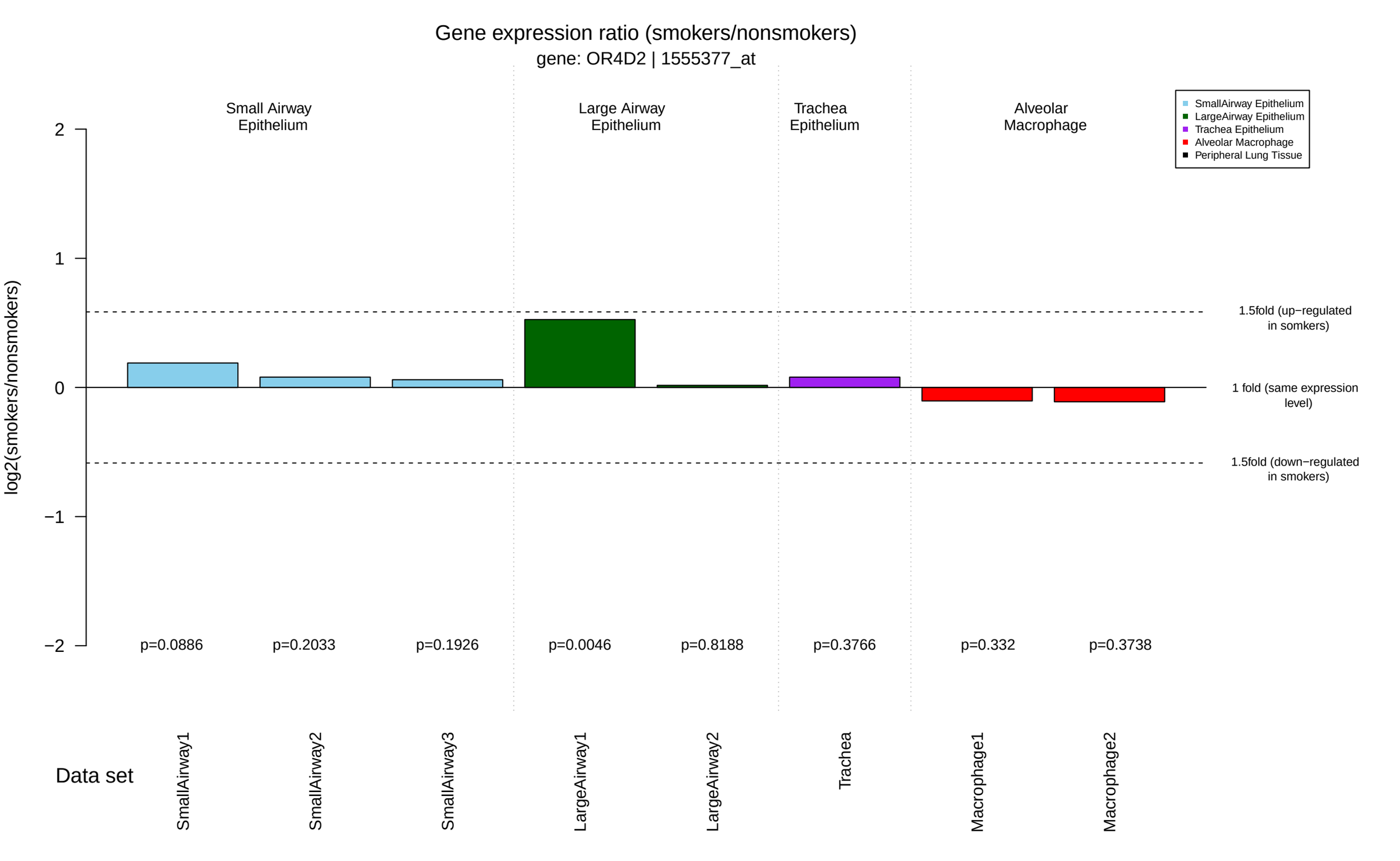
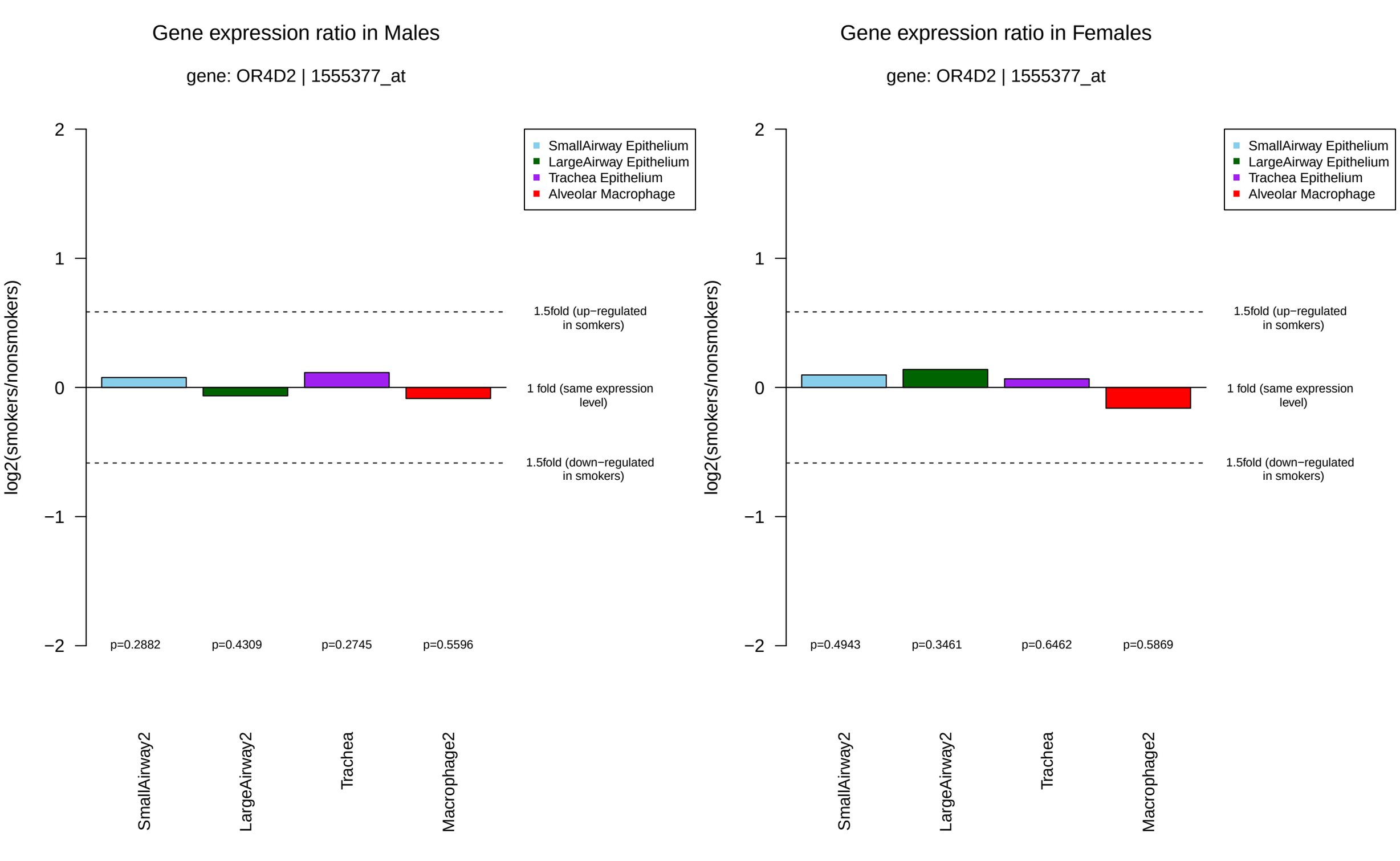
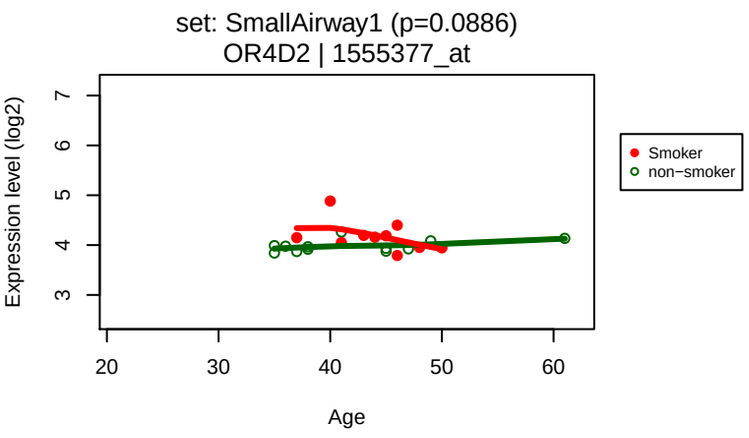
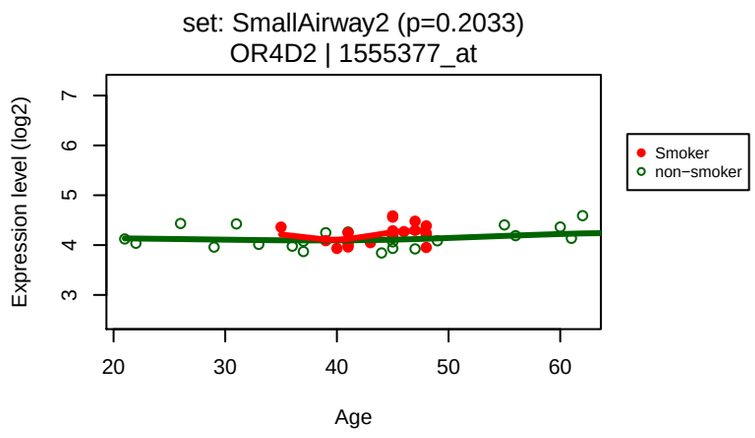
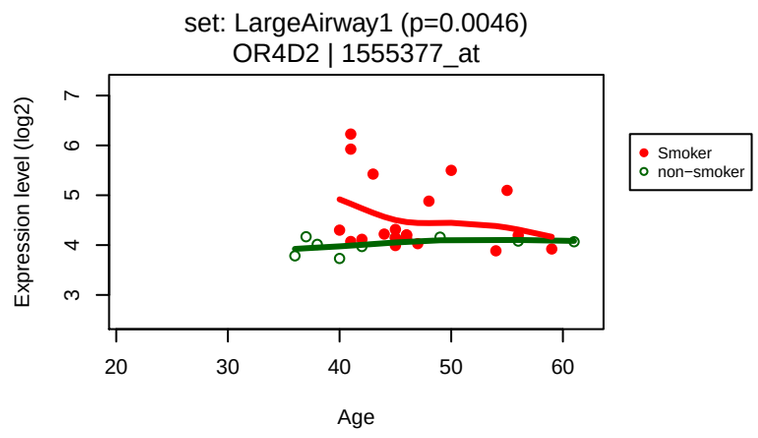

Gene expression of smokers and nonsmokers
gene: OR4D2 | 1555377_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0886
p=0.2033
p=0.1926
p=0.0046
p=0.8188
p=0.3766
p=0.332
p=0.3738
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

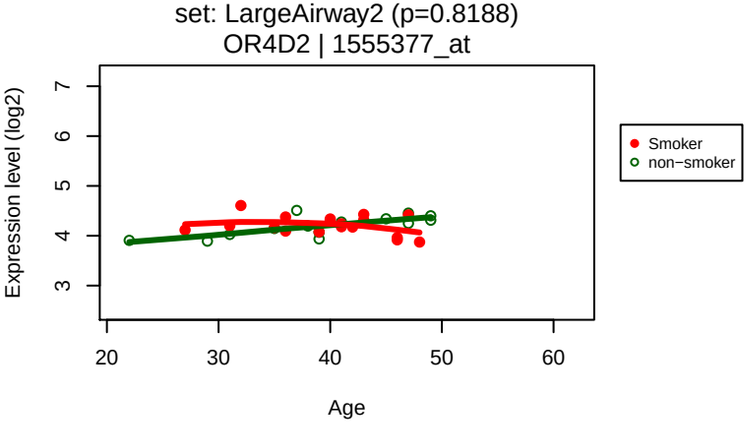

Gene expression ratio (smokers/nonsmokers)
gene: OR4D2 | 1555377_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0886
p=0.2033
p=0.1926
p=0.0046
p=0.8188
p=0.3766
p=0.332
p=0.3738
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

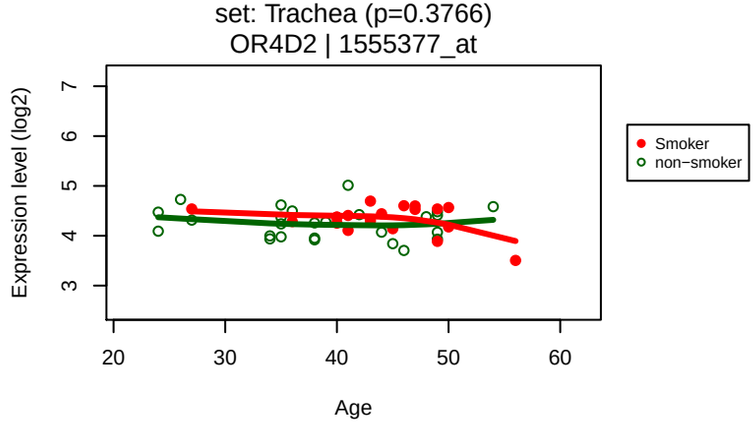

Gene expression ratio in Males
gene: OR4D2 | 1555377_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2882
p=0.4309
p=0.2745
p=0.5596
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR4D2 | 1555377_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4943
p=0.3461
p=0.6462
p=0.5869
SmallAirway2
LargeAirway2
Trachea
Macrophage2

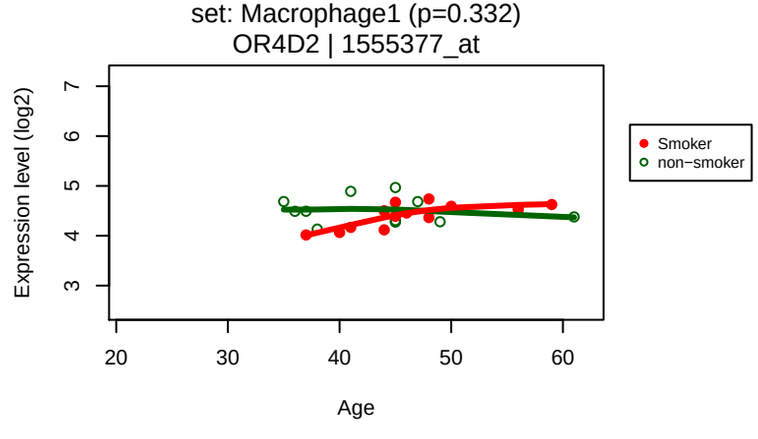

set: SmallAirway1 (p=0.0886)
OR4D2 | 1555377_at
Expression level (log2)
Age
Smoker
non−smoker

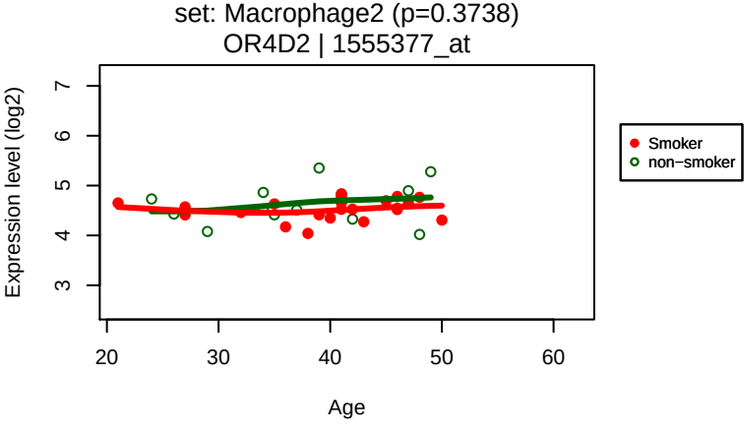

set: SmallAirway2 (p=0.2033)
OR4D2 | 1555377_at
Expression level (log2)
Age
Smoker
non−smoker

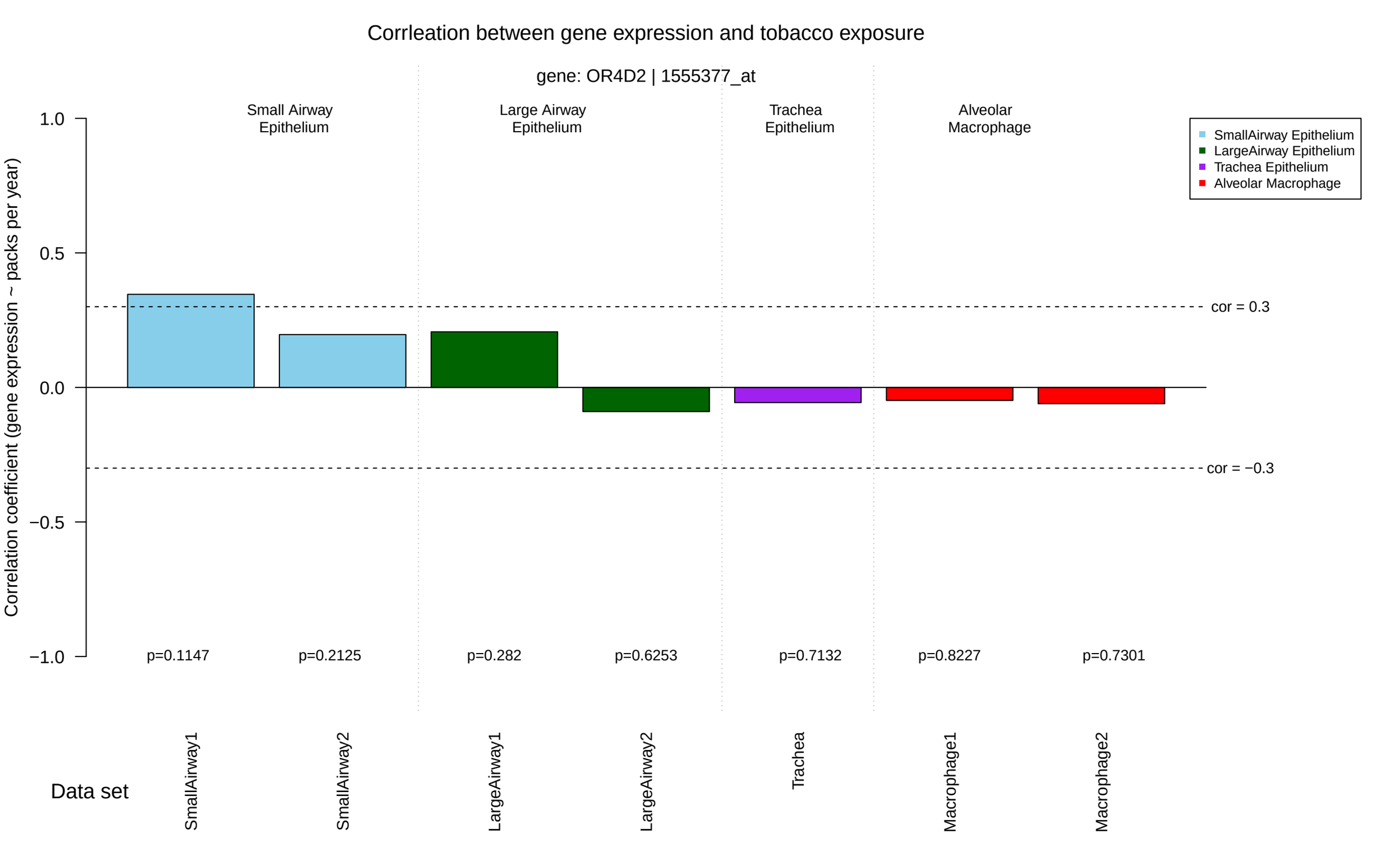

set: LargeAirway1 (p=0.0046)
OR4D2 | 1555377_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8188)
OR4D2 | 1555377_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3766)
OR4D2 | 1555377_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.332)
OR4D2 | 1555377_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3738)
OR4D2 | 1555377_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR4D2 | 1555377_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1147
p=0.2125
p=0.282
p=0.6253
p=0.7132
p=0.8227
p=0.7301
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2